

Leave Approval System Using ServiceNow

Project Documentation

1.Introduction:

The Leave Approval System is a workflow automation application developed using the ServiceNow platform to simplify and automate the employee leave management process. Instead of using manual methods such as emails or paper forms, the system enables employees to submit leave requests online.

2.Project Overview:

The Leave Approval System is a workflow automation application developed on the ServiceNow platform to simplify and streamline the employee leave management process. The system allows employees to submit leave requests by providing details such as leave type, start date, end date, and reason. It validates the leave dates to prevent invalid requests and automatically routes the application to the assigned manager using Flow Designer. Managers can review and approve or reject the request, after which the system updates the leave status and sends an email notification to the employee. By automating the entire leave approval process, the application reduces manual effort, improves accuracy, ensures transparency, and maintains a centralized record of all leave requests for efficient tracking and management.

3.Architecture:

The architecture of the Leave Approval System is organized into five layers that work together to automate the employee leave management process.

1. Users Layer

The Users Layer consists of two primary users:

Employee – Submits leave requests, views request status, and receives approval or rejection notifications.

Manager – Reviews leave requests and approves or rejects them based on organizational policies.

2. Presentation Layer

The Presentation Layer provides the user interface through the ServiceNow Web Portal. Employees and managers interact with the application using online forms to submit, review, and manage leave requests. The interface is designed to be user-friendly and accessible.

3. Application / Business Layer

This layer contains the core business logic of the Leave Approval System and is implemented using the ServiceNow platform.

Leave Request Application – Manages the creation and processing of leave requests.

Flow Designer (Approval Flow) – Automatically routes leave requests to the appropriate manager for approval.

Notifications (Email/In-App) – Sends automatic notifications to employees regarding approval or rejection decisions.

Business Rules & Validations – Validates input data, such as ensuring the end date is not earlier than the start date, and enforces business policies.

Reports & Dashboards – Generates reports and dashboards to monitor leave requests, approvals, and employee leave history.

4. Data Layer

The Data Layer stores all application data securely in the ServiceNow Database. It maintains information related to:

- **Leave Requests**
- **Leave Types**
- **Employee and Manager Details**
- **Approval Records**
- **Leave Status and History**

This layer ensures data integrity, reliability, and efficient retrieval of leave information.

5. Integration Layer

The Integration Layer connects the application with external services to enhance functionality.

Email Server – Sends approval and rejection notifications to employees.

User Directory (LDAP/Active Directory) – Authenticates users and retrieves employee information.

Calendar Integration (Outlook/Google Calendar) – Can synchronize approved leave with organizational calendars for better scheduling.

4.Setup Instructions:

Step 1: Create a Custom Application

- Navigate to **All > System Applications > Studio**
- Click Create Application
- Application Name: Leave Approval System
- Create the application.

Step 2: Create Leave Request Table

Create a table:

Table Name: Leave Request

Field Label	Type
Leave Number	Auto Number
Employee	Reference (User [sys_user])
Leave Type	Choice
Start Date	Date
End Date	Date
Reason	String / Journal
Manager	Reference (User [sys_user])
State	Choice

Leave Type Choices

- Casual Leave
- Sick Leave
- Earned Leave

- Work From Home

State Choices

- Requested
- Approved
- Rejected

Step 3: Prevent End Date Earlier Than Start Date

Create a **Client Script**

Type: onSubmit

Script:

```
function onSubmit() {  
  
    var start = new Date(g_form.getValue('start_date'));  
    var end = new Date(g_form.getValue('end_date'));  
  
    if (end < start) {  
        g_form.addErrorMessage(  
            'End Date cannot be earlier than Start Date');  
        return false;  
    }  
  
    return true;  
}
```

Step 4: Create Approval Flow

Navigate to: Flow Designer

Create Flow: Leave Request Approval

Trigger

When:

Leave Request record created

Condition:

State = Requested

Flow Steps

Step 1: Ask For Approval

Action: Ask For Approval

Approver: Manager field

Wait until approval decision.

Step 2: If Approved

Condition:

Approval = Approved

Update Record:

State = Approved

Step 3: If Rejected

Condition:

Approval = Rejected

Update Record:

State = Rejected

Step 6: Approval Notification

Navigate: **System Notification > Email > Notifications**

Create Notification: Approved Notification

Condition: State changes to Approved

Recipients: Employee

Subject: Leave Request Approved

Message:

Hello \${employee.name},

Your leave request from
\${start_date} to \${end_date}
has been approved.

Thank You.

Step 7: Rejection Notification

Create another notification.

Condition: State changes to Rejected

Recipients: Employee

Subject: Leave Request Rejected

Message:

Hello \${employee.name},

Your leave request from
\${start_date} to \${end_date}
has been rejected.

Thank You.

Step 8: Create Module/Menu

Create: Application Menu

Leave Approval System

Module: Apply Leave

Link: Leave Request table

Another Module: My Leave Requests

Filter: Employee = Me

5. Running the Application:

1. Log in to the ServiceNow instance.
2. Open the Leave Approval System application.
3. Navigate to Leave Requests and create a new leave request.
4. Enter employee details, leave type, start date, end date, and reason.
5. Submit the request.
6. The request is automatically routed to the assigned manager through Flow Designer.
7. The manager reviews and either Approves or Rejects the request.
8. The system updates the leave request status automatically.

9. An email notification is sent to the employee with the final decision.
10. The employee can view the updated status in the Leave Request record.